THE RANK DECOMPOSITION OF A FINITE LOCALLY FREE GROUPOID SCHEME INTO R-INVARIANT PIECES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lemma morphisms-lemma-finite-locally-free (canonical tag 04MH). cited at proof lines 4521-4526. Rank decomposition of the base.
- Lemma lemma-diagram (canonical tag 02YE). cited at proof lines 4528-4535. The fibre comparison behind R-invariance.

Source reference. Original lemma 04MH: sources/stacks-chow/tags.txt, source lines 6043–6043. Cited at proof lines 4521-4526 to produce the rank decomposition for s; load-bearing.

Source reference. Original lemma 02YE: sources/stacks-chow/tags.txt, source lines 3873–3873. Cited at proof lines 4528-4535 to compare the fibres of s over R-equivalent points via pr_1^{-1}(z); load-bearing.

Lemma 0.2. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid scheme over S. Assume $s, t : R \to U$ finite locally free. Then*

$$U = \coprod\nolimits_{r \geq 1} U_r$$

is a disjoint union of R-invariant opens such that the restriction R_r of R to U_r has the property that $s, t : R_r \to U_r$ are finite locally free of rank r.

Proof. By Morphisms, Lemma 04MH there exists a decomposition $U = \coprod_{r \geq 0} U_r$ such that $s : s^{-1}(U_r) \to U_r$ is finite locally free of rank r. As s is surjective we see that $U_0 = \emptyset$. Note that $u \in U_r \Leftrightarrow$ if and only if the scheme theoretic fibre $s^{-1}(u)$ has degree r over $\kappa(u)$. Now, if $z \in R$ with $s(z) = u$ and $t(z) = u'$ then using notation as in Lemma 02YE

$$\mathrm{pr}_1^{-1}(z) \to \mathrm{Spec}(\kappa(z))$$

is the base change of both $s^{-1}(u) \to \mathrm{Spec}(\kappa(u))$ and $s^{-1}(u') \to \mathrm{Spec}(\kappa(u'))$ by the lemma cited. Hence $u \in U_r \Leftrightarrow u' \in U_r$, in other words, the open subsets U_r are R-invariant. In particular the restriction of R to U_r is just $s^{-1}(U_r)$ and $s : R_r \to U_r$ is finite locally free of rank r. As $t : R_r \to U_r$ is isomorphic to s by the inverse of R_r we see that it has also rank r. □

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